18 Lecture 18 (February 26, 2025, est.): Products, equivariant objects, and tensor powers

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Throughout this lecture, k is algebraically closed. Our aim is to construct the Frobenius functor for a semisimple symmetric tensor category in characteristic p. We first make precise the product of categories and the category of objects equipped with a group action. These constructions will also explain the product decomposition of Ver_p from the previous lecture.

18.1 Products of linear categories

Let $\mathcal{C}$ and $\mathcal{D}$ be k-linear additive Karoubian categories. Begin with formal pairs $X \boxtimes Y$ and put

$$\mathrm{Hom}(X \boxtimes Y, X' \boxtimes Y') = \mathrm{Hom}_{\mathcal{C}}(X, X') \otimes_k \mathrm{Hom}_{\mathcal{D}}(Y, Y').$$

Composition is componentwise on elementary tensors. One must then take the additive hull, allowing finite formal direct sums, and its idempotent completion. We denote the resulting Karoubian product by $\mathcal{C} \boxtimes \mathcal{D}$ in this lecture. Merely taking pairs would not in general give an additive category.

If the factors are rigid symmetric monoidal, the formulas

$$(X \boxtimes Y) \otimes (X' \boxtimes Y') = (X \otimes X') \boxtimes (Y \otimes Y'), \qquad (X \boxtimes Y)^* = X^* \boxtimes Y^*$$

extend to give the product the same structures. There are canonical functors $X \mapsto X \boxtimes \mathbb{1}$ and $Y \mapsto \mathbb{1} \boxtimes Y$.

18.1 Proposition (Universal property)**.** *For a k-linear additive Karoubian symmetric monoidal category $\mathcal{A}$, giving an additive k-linear symmetric monoidal functor $\mathcal{C} \boxtimes \mathcal{D} \to \mathcal{A}$ is equivalent to giving one such functor from each factor to $\mathcal{A}$.*

Proof. Given $F : \mathcal{C} \to \mathcal{A}$ and $G : \mathcal{D} \to \mathcal{A}$, define $H(X \boxtimes Y) = F(X) \otimes G(Y)$ and $H(f \otimes g) = F(f) \otimes G(g)$. Its tensor constraint uses the symmetry to interchange the middle two factors:

$$F(X) \otimes G(Y) \otimes F(X') \otimes G(Y') \longrightarrow F(X \otimes X') \otimes G(Y \otimes Y').$$

The symmetry axioms give the required coherence. Additivity and splitting idempotents extend H to the completed product, uniquely up to natural isomorphism. Conversely, restrict a functor H to the two canonical copies of the factors. Since $X \boxtimes Y = (X \boxtimes \mathbb{1}) \otimes (\mathbb{1} \boxtimes Y)$, the two procedures are inverse, including on monoidal natural isomorphisms. ∎

18.2 Remark. This Karoubian product should not be confused with the general abelian Deligne tensor product, characterized by right-exact bilinear functors. The distinction disappears in the case used here, where one factor is split semisimple

and the other is abelian. If $\mathcal{C}$ is split semisimple with simple objects S, then, as a linear category,

$$\mathcal{C} \boxtimes \mathcal{D} \simeq \bigoplus_{S \in \mathrm{Irr}(\mathcal{C})} \mathcal{D},$$

where objects have finite support. Indeed, every object of the product is a finite sum of objects $S \boxtimes Y_S$, there are no morphisms between distinct S-blocks, and each block has exactly the morphisms of $\mathcal{D}$. Splitting idempotents preserves this description. In particular the product is abelian if $\mathcal{D}$ is. When both factors are split semisimple, its simples are precisely the pairs of simples.

18.3 Proposition. *Let $\mathcal{A}$ and $\mathcal{B}$ be full semisimple tensor subcategories of a semisimple symmetric tensor category $\mathcal{C}$. If their only common simple object is $\mathbb{1}$, then multiplication gives a symmetric equivalence*

$$\mathcal{A} \boxtimes \mathcal{B} \simeq \langle \mathcal{A}, \mathcal{B} \rangle,$$

where the right-hand side is the full tensor subcategory generated by the two subcategories.

Proof. For simple $X, X' \in \mathcal{A}$ and $Y, Y' \in \mathcal{B}$, rigidity and symmetry identify

$$\mathrm{Hom}(X \otimes Y, X' \otimes Y') \cong \mathrm{Hom}(X'^* \otimes X, Y' \otimes Y^*).$$

An object of $\mathcal{A}$ and an object of $\mathcal{B}$ can have only copies of the unit as common simple summands. The last Hom space is therefore the tensor product of the corresponding unit multiplicity spaces. Its dimension is 1 if $X \cong X'$ and $Y \cong Y'$, and is 0 otherwise. In particular, the $X \otimes Y$ are simple and distinct for distinct pairs. This proves full faithfulness of multiplication on the product, and its image is closed under sums and summands. Every tensor word in the two subcategories can be reordered to have all its $\mathcal{A}$ factors first, so the image is exactly the generated subcategory. The symmetry gives the monoidal structure as in the universal property above. ∎

For example, the subcategories SVec_k and Ver_p^+ inside Ver_p, for $p \geq 5$, have only the simple unit in common and generate the whole category. The proposition recovers their product decomposition.

18.2 Equivariant objects

An action of a group G on a category requires coherent functors, not just a homomorphism into the group of isomorphism classes of autoequivalences. Concretely, choose autoequivalences T_g and natural isomorphisms

$$\mu_{g,h} : T_g T_h \xrightarrow{\sim} T_{gh},$$

with a unit identification for T_1, satisfying the associativity cocycle identity. For a monoidal action, the T_g and the isomorphisms are monoidal.

18.4 Definition. An *equivariant object* is an object X together with isomorphisms $u_g : T_g(X) \to X$ satisfying

$$u_{gh} \circ \mu_{g,h,X} = u_g \circ T_g(u_h)$$

and the unit condition. A morphism $f : (X, u) \to (Y, v)$ satisfies $f \circ u_g = v_g \circ T_g(f)$ for all g. The resulting category is denoted $\mathcal{C}^G$.

For a monoidal action, tensor products of the maps u_g give $\mathcal{C}^G$ a monoidal structure. If the action respects a symmetry, the symmetry is equivariant as well. In the rest of this lecture the action is *trivial*: $T_g = \mathrm{id}$ and all coherence maps are identities. Then an equivariant object is simply (X, ρ) with $\rho : G \to \mathrm{Aut}_{\mathcal{C}}(X)$ a group homomorphism. For example, $\mathrm{Vect}_k^G = \mathrm{Rep}_k(G)$.

18.5 Proposition. *Let $\mathcal{C}$ be a semisimple symmetric tensor category and G a finite group acting trivially. There is a symmetric equivalence*

$$\mathcal{C}^G \simeq \mathcal{C} \boxtimes \mathrm{Rep}_k(G).$$

Proof. Write an object of $\mathcal{C}$ as $X = \bigoplus_S S \otimes_k M_S$, where the S are simple and the $M_S = \mathrm{Hom}(S, X)$ are finite-dimensional multiplicity spaces. The notation $S \otimes_k M_S$ means a direct sum of copies of S. Since $\mathrm{End}(S) = k$ and morphisms between distinct simples vanish, a G-action on X is exactly a G-representation on each M_S. Equivariant morphisms are exactly the equivariant linear maps on these spaces. This gives the claimed equivalence of categories. It is the functor sending $X \boxtimes V$ to $X \otimes_k V$ with G acting on V, and the componentwise tensor actions show that it is symmetric monoidal. ∎

18.3 Semisimplifying equivariant objects

18.6 Lemma. *A simple object S of a semisimple rigid symmetric category has nonzero categorical dimension.*

Proof. Rigidity gives $\dim_k \mathrm{Hom}(\mathbb{1}, S \otimes S^*) = \dim_k \mathrm{End}(S) = 1$, so $\mathbb{1}$ occurs with multiplicity one in $S \otimes S^*$. Both coevaluation and evaluation composed with the symmetry are nonzero, by the duality identities. In a semisimple category, a nonzero inclusion and a nonzero projection of a simple summand of multiplicity one have nonzero composite. This composite is $\dim(S)$. ∎

This assertion concerns simple objects. It does not say that all nonzero objects have nonzero dimension: in characteristic p, $\mathbb{1}^{\oplus p}$ has dimension zero.

18.7 Proposition. *In the situation of Proposition 18.5,*

$$\overline{\mathcal{C}^G} \simeq \mathcal{C} \boxtimes \overline{\mathrm{Rep}_k(G)}$$

as symmetric tensor categories.

Proof. The indecomposable objects of $\mathcal{C}^G$ are $S \boxtimes V$, where S is simple in $\mathcal{C}$ and V is an *indecomposable* representation of G; V need not be simple. The trace of an endomorphism of such an object is $\dim(S)$ times its ordinary trace on V. Thus nilpotent endomorphisms have trace zero, and

$$\dim(S \boxtimes V) = \dim(S)\dim_k(V).$$

By the lemma, this vanishes precisely when $\dim_k(V)$ vanishes in k. More generally, on each S-block the trace pairing on morphism spaces is the ordinary trace pairing on G-modules multiplied by the nonzero scalar $\dim(S)$. Its radical is therefore unchanged. Quotienting by these radicals gives exactly $\mathcal{C} \boxtimes \overline{\mathrm{Rep}_k(G)}$. Tensor products and symmetries descend from the equivalence before quotienting. ∎

18.4 The cyclic action on a tensor power

Assume now that $\operatorname{char} k = p > 0$ and that $\mathcal{C}$ is semisimple. The symmetry supplies an S_p-action on $X^{\otimes p}$. Restricting to the cycle $\sigma = (1\,2\,\cdots\,p)$ defines a functor

$$P : \mathcal{C} \longrightarrow \mathcal{C}^{C_p}, \qquad X \longmapsto (X^{\otimes p}, \sigma), \qquad f \longmapsto f^{\otimes p}.$$

Canonical rearrangement of tensor factors makes P symmetric monoidal. It is not additive. Compose it with semisimplification:

$$\mathrm{Fr}_0 : \mathcal{C} \xrightarrow{P} \mathcal{C}^{C_p} \longrightarrow \overline{\mathcal{C}^{C_p}} \simeq \mathcal{C} \boxtimes \mathrm{Ver}_p\,.$$

The essential point is that the mixed terms in a tensor power become negligible. In the expansion of $(X \oplus Y)^{\otimes p}$, the summands are indexed by words of length p in the letters X, Y. There are two constant words, giving $X^{\otimes p}$ and $Y^{\otimes p}$. Every other word has a free orbit under cyclic rotation because p is prime. Choose a summand W in such an orbit. Using the cyclic action to identify its translates, the sum of the orbit is the equivariant object $W \otimes_k k[C_p]$. Under the product description above, it is $W \boxtimes J_p$ and vanishes after semisimplification. Consequently the canonical inclusions and projections induce

$$\mathrm{Fr}_0(X \oplus Y) \cong \mathrm{Fr}_0(X) \oplus \mathrm{Fr}_0(Y).$$

The same orbit argument applied to morphisms proves additivity; we give that proof in the next lecture. Scalars satisfy

$$\mathrm{Fr}_0(\lambda f) = \lambda^p \mathrm{Fr}_0(f).$$

To express this as a k-linear functor, let $\mathcal{C}^{(1)}$ have the same underlying additive symmetric category as $\mathcal{C}$, but scalar action $\lambda \cdot_{(1)} f = \lambda^p f$. The field k is perfect, so this is again a category over k. Twist the whole target of Fr_0 in this manner, and use the fact that Ver_p is defined over $\mathbb{F}_p$ to identify its twist with itself. The resulting functor is

$$\mathrm{Fr} : \mathcal{C} \longrightarrow \mathcal{C}^{(1)} \boxtimes \mathrm{Ver}_p\,.$$

This is the Frobenius functor of [Ost15, Section 3.3]. After additivity is established, it is exact because every exact sequence in the source splits.

18.8 Remark. Semisimplicity is essential to the product description used in this construction. For a general symmetric tensor category one cannot obtain the correct Frobenius functor simply by taking the ordinary semisimplification of $\mathcal{C}^{C_p}$. The later lectures use a relative construction to treat that case. The questions of when a category admits a fiber functor to Ver_p, and why Ver_p cannot be further compressed by a tensor functor, will be addressed after studying this Frobenius functor.

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